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On the Application of Linear Programming on a Transportation Problem

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Abstract

This research demonstrates the application of Linear programming on a transportation problem to products of Lubcon limited. Data were collected as extracts from the records of the company on the five (5) products and six (6) districts offices. Information regarding the supply capacity and demand requirements were also given. We have been able to formulate a linear programming for the transportation problem for the products of Lubcon limited. We obtained the minimum transportation cost of #183,522,300 with some cartons of products to be transported to their destinations in order to attain a minimum cost which is the goal of the company.

Keywords: Linear programming, Simplex Method, Objective Function, Constraints, transportation problem

Introduction

Manufacturers are often faced with problems of moving or transporting their products or goods from sources to their various destinations due to the availability of limited resources. These resources are capital, manpower and the distances from the sources to the destinations. However, Network flow program is a program that describes a type of model that is a special case of the more general linear program. The class of network flow programs includes such problems as the transportation problems assignment problem, the shortest part problem, the maximum flow problem etc.

Linear programming (LP) can be defined as a mathematical technique for determining the best allocation of a firm's limited resources to achieve optimum goal. It is also a mathematical technique used in Operations Research (OR) or Management Sciences to solve specific types of problems such as allocation, transportation and assignment problems that permits a choice or choices between alternative courses of action (Yahya, 2004). Linear programming is a term that covers a whole range of mathematical techniques that is aimed at optimizing performance in terms of combinations of resources (Lucey, 1996).

Linear Programming being the most prominent OR technique, it is designed for models with linear objective and constraint functions. A LP model can be designed and solved to determine the best course of action as in a product mix subject to the available constraints.

The Simplex method, also called Simple technique or Simplex Algorithm, was invented by George Dantzig, an American Mathematician, in 1947. It is the basic workhorse for solving Linear Programming Problems up till today. There have been many refinements to the method, especially to take advantage of computer implementations, but the essentials elements are still the same as they were when the method was introduced (Chinneck, 2000; Gupta and Hira, 2006).

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Generally, the Objective function may be of maximization of profit or minimization of costs (which is the focus of this paper) or labor hours. Moreover, the model also consists of certain structural constraints which are set of conditions that the optimal solution should justify. Examples of the structural constraints include the raw material constraints, production time constraint, and skilled labor constraints to mention a few. An optimum solution is a solution that fulfills both the constraints of the problem and the set objective to be met.

In Balogun *et al* (2013b), Linear programming was used in maximizing profit and based on the analysis carried out in this research on Feed Master Limited, Ilorin the result showed that 25kg of Chick mash, 25kg of Layer mash, 25kg of Grower mash, 25kg of Broiler starter mash and 25kg of Broiler finisher mash should be produced but more of 25kg of Layer mash, 25kg of Grower mash and 25kg of Broiler finisher mash in order to satisfy their customers. Also, more of 25kg of Layer mash, 25kg of Grower mash and 25kg of Broiler finisher mash should be produced in order to attain maximum profit because they contribute mainly to the profit earned.

The term "linear", as stated by Akingbade (1996), implies proportionality, which means that the elements in a situation are so connected that they appear as straight line when graphed. While the "programming" indicates the solution method which can be carried out by an iterative process in which a researcher advance from one solution to better solution until a final solution is reached which cannot be improved upon. This final solution is termed the optimal solution of the LP problem. According to Modibbo et al, (2015) linear programming was used to optimize the foam produced by Bajabure Industrial Complex (Nima foam), Yola, Nigeria. The result showed that more of 25kg and 29kg density of the foam produced by the company in order to attain the Maximum profit the company desire.

Balogun *et al* (2013a), the researcher used Profit Maximization and based on the analysis carried out in this research using SOKAT Soap Industry, Ikotun shows that the company should produce 10kg, 5kg and 2kg of Salem soap but more of 1kg of Salem soap in order to satisfy their customers. Also, more of 1kg of Salem soap should be produced in order to attain maximum profit because it contributes mostly to the profit earned.

Yahya *et al* (2012), the work focuses on soap product manufactured by KASMO Industry Limited (KIL) Osogbo, Nigeria for Profit Maximization and the showed that the industry should produce 1 soap tablet, 3 soap tablet, 12 soap tablet and 120 soap tablet but more 1 soap tablet in order to optimize their profit.

Balogun *et al* (2012) carried out on Nigeria Bottling Company, Ilorin plant for Profit Maximization and the analysis revealed that Fanta Orange 50cl and 35cl, Coke 50cl and 35cl, Fanta lemon 35cl, Sprite 50cl, Schweppes, Krest soda 35cl should be produced but more of Coke 50cl and Fanta Orange 50cl in order to satisfy their customers. Also, more of Coke 50cl and Fanta Orange 50cl should be produced in order to attain maximum profit because they contribute usually to the profit earned.

The Transportation problem is a classic Operations Research problem where the objective is to determine the schedule for transporting goods from source to destination in a way that minimizes the shipping cost while satisfying supply and demand constraints (Blum and Roli, 2003).

The problem is to determine the quantity to be transported from each source to each demand center, so as to meet all the requirements at minimum total shipping cost (Winston, 1994).

Transportation problem deals with sources where a supply of some commodity is available and destinations where the commodity is demanded or needed. The classical statement of the transportation problem uses a matrix with the row representing sources and columns representing destinations.

The costs of shipping from sources to destination are indicated by the entries in the matrix and it is denoted by $C_{ij}\ where i = 1, 2, ..., m\ and\ j = 1, 2, ..., n.$ The problem then is to determine the quantity X_{ij} to be transported from all sources to all destinations so as to minimize the total cost of transportation.

Lubcon limited is a company that operates on a single shift located at Adewole Industrial Estate, Ilorin where their products (lubricants) are transported to their various district's offices in Aba, Abuja, Benin, Kano, Lagos and Maiduguri. The company is faced with the problem as regard their distribution objectives in that the distances of their various district offices from the headquarter (plant) are different and since the demand at their different district offices had been studied, it is therefore, necessary to improve on the transportation system because there is need to satisfy every destination. This necessitate the company to set-up their own well-organized transportation department who seek to overcome the transportation problem.

In general, the problems faced by manufacturers as regard their transportation system are capital, manpower and distances from the sources to their destinations.

Specifically, the objective of Lubcon limited is to transport their quality products from headquarter in Ilorin to all destinations (district offices) where the products are needed by its customers at a minimum cost.

Material and Methods

The dataset used for this work was collected as extracts from records of Lubcon limited, Ilorin, Kwara State, Nigeria on products transported to all their district offices at a minimum cost.

The analysis was carried out using linear programming techniques. The linear programming problem developed here is a mathematical program in which the objective function is linear in the unknown variables and the constraints have linear equation or linear inequality or both. The general form of a linear programming problem is stated in matrix form as show below:

$$Minimize \;\; Z = \sum_{j=1}^{r} C_j X_j \qquad (1)$$

$$Subject \;\; to \;\; \sum_{j=1}^{r} a_{ij} X_j \le b_i, \; (b_i \ge 0), \; i = 1,2,3,...,m \qquad (2)$$

$$X_j \ge 0, \;\; j = 1,2,3,...,m \qquad (3)$$

The C in (1) is a row vector of m-dimension representing the objective function coefficients, X is a $m \times 1$ column vector of the decision variable of the LP model, A is a $m \times k$ matrix of coefficients, and b is a $k \times 1$ column vector of values in the right hand side of the constraint equations in (2) (Yahya *et al*, 2012). POM for windows software was used to carry out the analysis

Data Analysis

The data collected from Lubcon Limited, Ilorin, Nigeria on the goods transported from her headquarters to all her district office were analyzed to determine the how best to transport her goods in order minimize cost.

Minimize the objective function (costs):

$$\begin{aligned} Z = {} & 150X_{11} + 120X_{12} + 68X_{13} + 72X_{14} + 100X_{15} + 45X_{16} + 20X_{21} + 38X_{22} + 14X_{23} + 18X_{24} + 40X_{25} + 10X_{26} \\ & + 90X_{31} + 51X_{32} + 33X_{33} + 42X_{34} + 69X_{35} + 44X_{36} + 30X_{41} + 48X_{42} + 24X_{43} + 16X_{44} + 28X_{45} \\ & + 5X_{46} + 15X_{51} + 27X_{52} + 13X_{53} + 12X_{54} + 18X_{55} + 4X_{56} \end{aligned}$$

$$\begin{aligned} & Subject\ to \\ & X_{11} + X_{12} + X_{13} + X_{14} + X_{15} + X_{16} = 44.00 \\ & X_{21} + X_{22} + X_{23} + X_{24} + X_{25} + X_{26} = 00.34 \\ & X_{31} + X_{32} + X_{33} + X_{34} + X_{35} + X_{36} = 54.95 \\ & X_{41} + X_{42} + X_{43} + X_{44} + X_{45} + X_{46} = 28.00 \\ & X_{51} + X_{52} + X_{53} + X_{54} + X_{55} + X_{56} = 07.47 \\ & X_{11} + X_{21} + X_{31} + X_{41} + X_{51} = 47.61 \\ & X_{12} + X_{22} + X_{32} + X_{42} + X_{52} = 48.18 \\ & X_{13} + X_{23} + X_{33} + X_{43} + X_{53} = 00.19 \\ & X_{14} + X_{24} + X_{34} + X_{44} + X_{54} = 08.67 \\ & X_{15} + X_{25} + X_{35} + X_{45} + X_{55} = 28.82 \\ & X_{16} + X_{26} + X_{36} + X_{46} + X_{56} = 00.29 \end{aligned}$$

With non-negativity constraint that $X_{ij} \ge 0, i = 1,2,3,4,5 and j = 1,2,3,4,5,6$

For analysis proper, the Simplex method proposed by George B. Dantzig as published in (Dantzig, 1963) was adopted to solve the above LP problem. The Simplex method has been found to be more efficient and convenient for computer software implementation in many instances (Yahya, 2004).

Results

The results from the analysis carried out on the LP model in the previous section using Simplex method estimated the value of the objective function to be 183,522,300. The contribution of the goods to transported to their destinations (district offices) as follows:

$X_{16} = 44, X_{21} = 0.34, X_{32} = 48.18, X_{33} = 0.19, X_{34} = 6.58, X_{44} = 2.09, X_{46} = 25.91, X_{51} = 47.27, X_{55} = 28.82$ respectively. This shows only variables mentioned above contributed meaningfully to improve the value of the objective function of the LP model.

Discussion and Conclusion

The appropriateness of the linear programming methods for minimum cost in the company has been demonstrated in this research. This is evident from the result obtained from the cost minimization type of the LP model fitted to the data collected on Lubcon Limited, Ilorin, Nigeria.

From the results of the LP model, it is desirable for Lubcon Limited, Ilorin, Nigeria to concentrate on the transportation of the following:

1,320,000 cartons of Adrenalin to Maiduguri

10,200 cartons of Dieselube to Aba

1,445,400 cartons of Lubcon Supper to Abuja

5,700 cartons of Lubcon Supper to Benin

197,000 cartons of Lubcon Supper to Kano

62,000 cartons of Nata Power to Kano

777,300 cartons of Nata Power to Maiduguri

1,418,100 cartons of Nata Supper to Aba

864,600 cartons of Nata Supper to Lagos

This would fetch the company a minimum cost of about #183,522,300 per month based on the goods transported to their destinations (district offices).

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